

MFT

(1) _____ layer of the OSI model is responsible for error-free data transmission

- a) Physical Layer
- b) Data Link Layer
- c) Network Layer
- d) Transport Layer

(2) The function of the Data Link Layer is _____

- a) Provides logical addressing and routing
- b) Establishes, maintains, and terminates connections between applications
- c) Ensures error-free transmission of data
- d) Provides a means for devices to access and transmit data over the physical layer

(3)The main protocol used in the Data Link Layer is _____

- a) TCP
- b) IP
- c) Ethernet
- d) HTTP

(4) The difference between a MAC address and an IP address is _____

- a) MAC addresses are physical addresses, while IP addresses are logical addresses.
- b) MAC addresses are used for routing, while IP addresses are used for identifying devices on a network.
- c) MAC addresses are unique to each device, while IP addresses are assigned to devices by a network administrator.
- d) MAC addresses are used in the Data Link Layer, while IP addresses are used in the Network Layer.

(5) One of the following is not a function of the Data Link Layer

- a) Error detection and correction
- b) Framing

- c) Flow control
- d) Logical addressing

(6) The purpose of a checksum in the Data Link Layer is _____

- a) To detect errors in the data
- b) To identify the source and destination of the data
- c) To regulate the flow of data between devices
- d) To provide security for the data

(7) The maximum frame size for Ethernet is

- a) 64 bytes
- b) 128 bytes
- c) 256 bytes
- d) 1500 bytes

(8) Layer of the OSI model is responsible for network interface communication

- a) Physical layer
- b) Data Link layer
- c) Network layer
- d) Transport layer

(9) _____ is a type of network interface

- a) Ethernet
- b) USB
- c) HDMI
- d) VGA

(10) The purpose of the MAC address in a network interface is

- a) To identify the manufacturer of the network interface
- b) To identify the IP address of the network interface

- c) To identify the physical location of the network interface
- d) To identify the network interface itself

Long Answer

- (1) Explain the term checksum in reference to error detecting codes.
- (2) Explain Physical addressing? Discuss the purpose of physical address?
- (3) Compare and contrast the different types of Data Link layer protocols, including their strengths and weaknesses.
- (4) Describe the process of error detection and correction in the Data Link layer. How do different protocols handle errors?
- (5) Explain the concept of framing in the Data Link layer. What is the purpose of framing, and how is it accomplished?
- (6) Explain the working of network interfaces.
- (7) Discuss the various types of Data Link layer addressing, including unicast, multicast, and broadcast. List the benefits and drawbacks of each type?
- (8) Write short notes on following:
 - a) ARP Protocol
 - b) Sliding Window Protocol
 - c) RARP Protocol
- (9) Explain TCP and UDP ALONG with packet format.
- (10) a) Differentiate between Stop and Wait & Go Back N sliding window protocol.
 - b) Briefly discuss about data link layer design issues?
- (11) Discuss each command in detail used in networking:
 - a) ipconfig
 - b) netstat
 - c) ping
 - d) hostname
 - e) tracer
- (12) A bit stream 10011101 is transmitted using $x^3 + 1$ generator polynomial. Generate the CRC code word for this message.

(13) Discuss the issues in the data link layer and about its protocol on the basis of layering principle.

(14) Critique the effectiveness of the Address Resolution Protocol (ARP) and explain any limitations of its use.

(15) A message that is to be transmitted is represented by the polynomial $M(x) = x^5 + x^4 + x$ with a generating prime polynomial $G(x) = x^3 + x^2 + 1$. Generate a 3 bit CRC code, $C(x)$ which is to be appended to $M(x)$.

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MFT

(1) _____ layer of the OSI model is responsible for routing data between different networks?

- a) Physical layer
- b) Data Link layer
- c) Network layer
- d) Transport layer

(2) _____ protocol is used by the network layer for routing data between networks?

- a) TCP
- b) UDP
- c) IP
- d) HTTP

(3) Addressing scheme used by the IP protocol is _____

- a) MAC addresses

- b) IP addresses
- c) URL addresses
- d) DNS addresses

(4) The purpose of the ARP protocol is _____

- a) To resolve IP addresses to MAC addresses
- b) To resolve MAC addresses to IP addresses
- c) To resolve URL addresses to IP addresses
- d) To resolve DNS addresses to IP addresses

(5) _____ is not a valid IP address

- a) 192.168.0.1
- b) 10.0.0.256
- c) 172.16.0.0
- d) 255.255.255.0

(6) _____ of the following is a private IP address

- a) 192.168.1.1
- b) 172.31.255.255
- c) 10.0.0.1
- d) All of the above

(7) _____ protocol is used by routers to exchange information about network topology and routes

- a) ICMP
- b) OSPF
- c) FTP
- d) HTTP

(8) _____ is not a valid routing algorithm

- a) Shortest Path First (SPF)

b) Distance Vector Routing

c) Link State Routing

d) Round Robin Routing

(9) The purpose of the TTL field in an IP packet is _____

a) To prevent routing loops

b) To specify the priority of the packet

c) To specify the length of the packet

d) To specify the destination network

(10) _____ is not a valid IP protocol

a) TCP

b) UDP

c) ICMP

d) HTTP

Long Answer

(11) Discuss the role of the network layer in the OSI model?

(12) Differentiate between the MAC address and the IP address.

(13) Explain routing information protocol(RIP) with working example in detail.

(14) Discuss how does the network layer ensure that data is transmitted to the correct destination?

(15) Explain the IPv4 datagram format in detail.

(16) Explain the IPv6 datagram format. Discuss the need for IPv6 addressing.

(17) Explain IP addressing. Discuss its classification? Describe how subnet addressing is performed?

(18) Compare and contrast IPv4 and IPv6, including their addressing schemes and any limitations or benefits.

(19) Analyze the different types of attacks that can be launched against the network layer, including denial-of-service (DoS) attacks and IP spoofing.

(20) Write short notes on:

a) ICMP Protocol

b) Net id

c) Host id

(21) Write short notes on:

a) OSPF

b) Tunneling

(22) a) Describe in detail Network Address Translation (NAT).

b) Class B network on the Internet has a subnet mask of 255.255.240.0. calculate the maximum number of hosts per subnet?

(23) Change the following IPv4 addresses from binary notation to dotted-decimal notation.

a) 10000001 00001011 00001011 11101111

b) 11000001 10000011 00011011 11111111

(24) Change the following IPv4 addresses from dotted-decimal notation to binary notation.

a) 111.56.45.78

b) 221.34.7.82

(25) A block of addresses is granted to a small organization. We know that one of the addresses is 205.16.37.39/28.

a) Find the first address in the block?

b) Find the last address for the block

c) Find the number of addresses

(26) Find the sub network address and the host-ID for the following

a) IP Address – 120.14.22.16 & Mask- 255.255.128.0

b) IP Address – 140.11.36.22 & Mask- 255.255.255.0

c) IP Address – 141.181.14.16 & Mask- 255.255.224.0

d) IP Address – 200.34.22.156 & Mask- 255.255.255.240

MFT

(1)_____ layer of the OSI model is responsible for end-to-end communication?

- a) Physical layer
- b) Data Link layer
- c) Transport layer
- d) Network layer

(2)_____ protocol is used by the Transport layer to provide reliable communication

- a) TCP
- b) UDP
- c) ICMP
- d) HTTP

(3) _____ Transport layer protocol is connectionless and provides no error checking

- a) TCP
- b) UDP
- c) ICMP
- d) HTTP

(4)_____ is considered to be more reliable,

- a) TCP
- b) UDP
- c) both a & b
- d) none of the above

5)_____ Transport layer protocol is typically used for streaming media and video conferencing

- a) TCP

- b) UDP
- c) both a & b
- d) none of the above

6) Which Transport layer protocol is used for email transmission?

- a. TCP
- b. UDP
- c. SMTP
- d. FTP

(7) The purpose of the Transmission Control Protocol (TCP) in the Transport layer is _____

- a) To provide error-free delivery of data
- b) To provide fast delivery of data
- c) To provide connectionless communication
- d) To provide congestion control

(8) The purpose of the User Datagram Protocol (UDP) in the Transport layer is

- a) To provide error-free delivery of data
- b) To provide fast delivery of data
- c) To provide connectionless communication
- d) To provide congestion control

(9) _____ is flow control in the Transport layer, and why is it necessary

- a) To prevent congestion in the network
- b) To prevent data loss due to buffer overflow
- c) To ensure timely delivery of data
- d) To ensure error-free delivery of data

(10) An endpoint of an inter-process communication flow across a computer network is called _____

- a) socket
- b) pipe
- c) port
- d) machine

Long Answer

(11) Discuss the role of the Transport layer in the OSI model and explain its importance.

(12) Compare and contrast the Transmission Control Protocol (TCP) and User Datagram Protocol (UDP) in terms of their reliability, speed, and use cases.

(13) Explain how the Transport layer provides error detection and correction, and give an example of how this is achieved in TCP.

(14) Describe the three-way handshake process used by TCP to establish a connection, and explain why it is necessary.

(15) Explain the concept of flow control in the Transport layer, and describe how it is achieved in both TCP and UDP.

(16) Explain the congestion control techniques used to improve QoS of the computer network.

(17) Describe the role of the port number in the Transport layer, and give an example of how it is used to distinguish between different applications.

(18) Explain how multiplexing and demultiplexing are used in the Transport layer to allow multiple applications to communicate over a single network connection.

(19) Describe the role of the Transport layer in Quality of Service (QoS) and how it can be used to prioritize different types of traffic.

(20) Explain the difference between connection-oriented and connectionless communication in the Transport layer, and give an example of a protocol that uses each type of communication.

(21) Describe the three-way handshake process used by TCP for transferring the data.

(22) Explain the TCP Segment format in detail.

(23) Explain the UDP header format in detail.

(24) The UDP header in hexadecimal format is as : 0632000D001CE217

Obtain the following from it:

- a) Source port number
- b) Destination port number
- c) Total length
- d) Length of the data.

(25) Explain how the transport layer ensure that the complete message arrives at the destination and in the proper order.

MFT

(1) _____ layer of the OSI model is responsible for providing services to end-users?

- a) Physical layer
- b) Data Link layer
- c) Transport layer
- d) Application layer

(2) _____ protocol is used by the Application layer to transfer files between hosts?

- a) SMTP
- b) FTP
- c) HTTP
- d) DNS

(3) _____ is used by the Application layer for sending and receiving email?

- a) SMTP
- b) FTP
- c) HTTP
- d) DNS

(4) _____ protocol is used by the Application layer for resolving domain names to IP addresses?

- a) SMTP
- b) FTP
- c) HTTP
- d) DNS

(5) _____ protocol is used by the Application layer for accessing web pages?

- a) SMTP
- b) FTP

c)HTTP

d)DNS

(6) _____ Application layer protocol is used for real-time communication, such as video conferencing?

a) SIP

b) SNMP

c) RTP

d) SSH

(7) The purpose of the Simple Mail Transfer Protocol (SMTP) in the Application layer is _____

a) To transfer files between hosts

b) To send and receive email

c) To resolve domain names to IP addresses

d) To access web pages

(8) The purpose of the Hypertext Transfer Protocol (HTTP) in the Application layer is _____

a) To transfer files between hosts

b) To send and receive email

c) To resolve domain names to IP addresses

d) To access web pages

(9) _____ Application layer protocol is used for remote login and command execution

a) SSH

b) SNMP

c) DNS

d) FTP

(10) Application layer protocol defines _____

a) types of messages exchanged

b) message format, syntax and semantics

c) rules for when and how processes send and respond to messages

d) all of the mentioned

Long Answer

- (1) Explain the importance of cryptography .Discuss in detail two types of cryptography.
- (2) Why is an application such as POP needed for electronic messaging?
- (3) Differentiate between HTTP,FTP and SMTP.
- (4) Elaborate the wireless connectivity in detail.
- (5) Explain firewall and its types.
- (6) With the help of diagram explain Zigbee architecture in detail.
- (7) Write briefly about World wide web. Also write down the three types of WWW documents.
- (8) Explain the difference between the public key and private key in Cryptography.
- (9) Explain the digital signature process in detail ?
- (10) Explain about Application layer and its services in detail.
- (11) Explain how security is provided in internet operations in detail
- (12) Differentiate between lossy and lossless compression techniques.
- (13) Differentiate between Bluetooth and Infrared.
- (14) Describe the role of a DNS on a computer network with reference to its components.
- (15) Discuss the features of HTTP and also discuss how HTTP works.
- (16) Explain the working of Electronic mail. How SMTP used in Email applications
- (17) Write short notes on any two of the following:
 - a) TELNET
 - b) BLUETOOTH
 - c) IEEE802.16 WIMAX
- (18) Differentiate between lossy and lossless compression techniques.
- (19) Discuss the design issues of application layer.

(20) List and explain various data compression techniques.